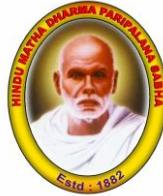


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SEMINAR ABSTRACT

**INTERPRETABLE DEEP LEARNING MODELS FOR
CREDIT RISK/LOAN APPROVAL (XAI)**

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ABSTRACT

Credit scoring and loan prediction are fundamental tasks in the financial sector, enabling institutions to assess borrower risk while ensuring efficient and reliable lending decisions. Recent advancements in Machine Learning (ML), Deep Learning (DL), and Explainable Artificial Intelligence (XAI) have significantly improved the accuracy, fairness, and transparency of credit risk assessment. This study presents a comprehensive review of four recent IEEE research papers that explore ensemble learning, deep learning architectures, fairness-aware AI, and explainable models for credit scoring and loan prediction. The reviewed works employ advanced techniques such as Random Forest, XGBoost, LightGBM, Long Short-Term Memory (LSTM), Transformer-based models, DenseNet, ResNet, SHAP, and data balancing methods including SMOTE and SMOTE-ENN to address challenges such as class imbalance, model interpretability, and prediction accuracy. Furthermore, the studies highlight the growing importance of fairness-aware learning, regulatory compliance, and ethical AI in financial decision-making. A comparative analysis of these approaches demonstrates that integrating ensemble and deep learning models with explainability and fairness techniques results in more robust, accurate, and trustworthy credit assessment systems. The review also identifies current research gaps, including the need for multimodal data integration, enhanced interpretability, and bias mitigation, providing valuable insights for future research in intelligent financial risk prediction systems.

Keywords :- Credit Scoring, Loan Prediction, Credit Risk Assessment, Machine Learning, Deep Learning, Explainable Artificial Intelligence (XAI), SHAP, Ensemble Learning, XGBoost, LightGBM, LSTM, Transformer, Fairness-Aware AI, SMOTE, Financial Risk Prediction.